



Test Project

54 Cyber Security

Module A1 Assessing Security



**BAGONG
PILIPINAS**



**Sa TESDA,
kayangKaya**



Contents

Introduction to Test Project 3

Introduction..... 3

Physical Topology..... 3

Logical Topology 4

Description of Project and Tasks 5

Target Machine Scanning Task..... 5

CMS Vulnerability Assessment Task..... 6

System Security Weaknesses Task..... 7

Analysis and Report Task 8

Instructions to the Competitor 10

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

Introduction to Test Project

You have been assigned by your company’s IT department to perform a security assessment and a basic penetration test on a website server running a Content Management System (CMS) that has been repeatedly targeted by hackers. As a cybersecurity expert, your responsibilities include evaluating the system’s security, documenting identified vulnerabilities, addressing key questions, and delivering an executive summary with an assessment of security risks and mitigation recommendations, along with any other relevant information as appropriate.

Login credentials for the competitor Laptops are as follows:

User: **competitor1a**

Password: **Boracay@14!**

Introduction

The competitors from each team will use VMWare Workstation to access VMs located at each competitor’s laptop. Each competitor will have the same configuration and VMs, but only have access to his/her own VMs.

The logical topology of the Virtual Machines at each competitor’s laptop is provided as follows:



Instructions to the Competitor

The VMs should all be preset with the correct IP addresses, subnet masks, and gateways, but it is the Competitor’s responsibility to verify or identify any errors or discrepancy between the VMs provided and this documentation, and to correct them if necessary.

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

14th WorldSkills ASEAN Competition • Manila, Philippines

Version 1.0 • Updated on 27 January 2025
Adopted & Adapted from WorldSkills Lyon 2024

Table 1. VM Finished Configuration Summary

VIRTUAL MACHINE	IP ADDRESS	NOTES	CREDENTIALS
Linux Server with CMS	192.168.2.1	Target machine for assessment	N/A
Kali Linux	192.168.2.2	Vulnerability assessment machine given to the assessor	kali/kali

Examine and familiarize yourself with the Virtual Machines, answer the required questions, and provide an executive summary, security risk assessment, mitigation recommendations, as well as any other relevant information as appropriate. The answer to the questions at each task should be supported by screenshots. The supporting screenshot should capture the key steps to obtain the answer, e.g. output of an exploitation command/tool,

This document will be provided digitally on the competitor’s workstation. Combine all answer documents into a single file for submission and save it on the competitor’s desktop. Once you signal to the experts that you have completed the task, the file will be collected. Please ensure that the file is named using your country code, and that the experts are able to retrieve it before you leave the competition venue.

Description of Project and Tasks

You have been provided with a Kali Linux machine to perform a vulnerability assessment and a basic penetration test on a Linux server running a CMS. You should begin by scanning and examining the services that may present potential vulnerabilities enabled on the system. Next, you are required to investigate CMS-related vulnerabilities, which could lead to system access or privilege escalation due to user accounts that do not comply with the system’s security policies. You are requested to answer the questions given and provide a report on the results of the vulnerability assessment and penetration testing according to the specified requirements below.

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

14th WorldSkills ASEAN Competition • Manila, Philippines

Version 1.0 • Updated on 27 January 2025
Adopted & Adapted from WorldSkills Lyon 2024

Information Gathering Task

Your initial task is to scan the target server using the provided IP address to identify services that may be vulnerable or serve as potential attack vectors for hackers, and then answer the following questions.

Table 1. Questions of Information Gathering Task

No	Task and Question	Answer
1	Identify the services running on the target server.	
	(Supporting Screenshot)	
2	Attempt to connect to one of the services of the target server, then search for and uncover the sensitive secret message within that service.	
	(Supporting Screenshot)	

CMS Vulnerability Assessment Task

The CMS vulnerability assessment task requires you to examine the basic information of the CMS running on the target server, then attempt an initial penetration test to identify potential vulnerabilities, and answer the specified questions as follows.

Table 2. Questions of CMS Vulnerability Assessment Task

No	Task and Question	Answer
1	Identify the version number of the CMS service that is currently running on the target server.	

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

14th WorldSkills ASEAN Competition • Manila, Philippines

Version 1.0 • Updated on 27 January 2025
Adopted & Adapted from WorldSkills Lyon 2024

No	Task and Question	Answer
	(Supporting Screenshot)	
2	Perform a penetration test on the CMS service to evaluate its vulnerabilities, and attempt to uncover and identify sensitive information stored within the system.	
	(Supporting Screenshot)	

System Security Weaknesses Task

In this task, you are required to identify security weaknesses within the system’s configuration. Suspicious activity has been detected from a user account on the CMS service, which appears to violate security policies and may provide an entry point for attackers to gain access to the server’s internal resources. Your objective is to locate and analyze the account in question, then respond to the following questions in the designated sections.

Table 3. Questions of System Security Weaknesses Task

No	Task and Question	Answer
1	Identify the user account on the CMS service that exposes the system weakness. Indicate the username as the answer.	
	(Supporting Screenshot)	

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

14th WorldSkills ASEAN Competition • Manila, Philippines

Version 1.0 • Updated on 27 January 2025
Adopted & Adapted from WorldSkills Lyon 2024

No	Task and Question	Answer
2	Conduct a penetration test to identify the password associated with this user account.	
	(Supporting Screenshot)	
3	Locate and identify sensitive information stored within the vulnerable user account’s home directory on the target machine.	
	(Supporting Screenshot)	
4	Perform a penetration test to gain root access on the target machine and collect sensitive information from the root account directory.	
	(Supporting Screenshot)	

Analysis and Report Task

The final task requires you to analyze the findings and compile a short report that clearly presents the results of your assessment. The report should begin with an executive summary (in less than 150 words) highlighting the identified vulnerabilities and security weaknesses, followed by an evaluation of the associated security risks and practical recommendations. You should specify the top 3 security risks that were identified during the assessment. The report should be prepared in accordance with the provided template.

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)

14th WorldSkills ASEAN Competition • Manila, Philippines

Version 1.0 • Updated on 27 January 2025
Adopted & Adapted from WorldSkills Lyon 2024

Table 4. Template of Analysis and Report Task

1. Executive Summary

2. Security Risk and Mitigation Recommendation		
Description	Security Risk (Low / Medium / High / Critical)	Mitigation Recommendation

Country 1 (CE)	Country 2 (DCE)	Country 3 (E)	Country 4 (E)	Country 5 (E)
Country 6 (E)	Country 7 (E)	Country 8 (E)	Country 9 (E)	Country 10 (E)